

Factorising

? ()

Factorising is the reverse of expanding brackets. We need to put our expression back into brackets.

We need to identify the common factors between both terms. These could be numbers, letters, or both.

Select the highest common factor and place this on the outside of the brackets.

To fill the inside of the brackets, ask yourself: "What do I need to multiply the common factor by to obtain the expression I started with?"

$$6a + 12$$

1, 2, 3, 6

6()

$$6 \times ? = 6a$$

$$6 \times ? = 12$$

$$6(a + 2)$$

$$4x^2 + 8x$$

x , 2, 4

This time we have a letter and a number.

$4x$ ()

$$4x \times ? = 4x^2$$

$$4x \times ? = 8x$$

$$4x(x + 2)$$